

MAT 322 HW09

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Problem 1. For each of the following functions $T: \mathbb{R}^4 \times \mathbb{R}^4 \rightarrow \mathbb{R}$, verify that T is a 2-tensor on $\mathbb{R}^4$ and express T in terms of the standard basis of $\mathcal{L}^2(\mathbb{R}^4)$.

(a) $T(x, y) = x_1y_3 + 2x_3y_1 - 3x_2y_4$

(b) $T(x, y) = 7x_4y_2 + 2x_2y_3 + 11x_1y_1$

Solution. We note that the basis elements $e_i^* \otimes e_j^*$ are the tensors such that $(e_i^* \otimes e_j^*)(x, y) = x_iy_j$, so for (a) we can write

$$T = e_1^* \otimes e_3^* + 2e_3^* \otimes e_1^* - 3e_2^* \otimes e_4^*$$

and for (b) we can write

$$T = 7e_4^* \otimes e_2^* + 2e_2^* \otimes e_3^* + 11e_1^* \otimes e_1^*.$$

In either case, we can see that T must be a 2-tensor on $\mathbb{R}^4$ because it is a linear combination of 2-tensors on $\mathbb{R}^4$. $\square$

Problem 2. For each of the following functions $\omega: \mathbb{R}^4 \times \mathbb{R}^4 \rightarrow \mathbb{R}$, verify that ω is an alternating 2-tensor on $\mathbb{R}^4$ and express ω in terms of the standard basis of $\mathcal{A}^2(\mathbb{R}^4)$.

(a) $\omega(x, y) = x_1y_3 - y_1x_3 + 2x_3y_1 - 2y_3x_1 - 3x_2y_3 + 3y_2x_3$

(b) $\omega(x, y) = 7x_1y_2 - 7y_1x_2 + 2x_2y_3 - 2y_2x_3$

Solution. We can write $(e_i^* \wedge e_j^*)(x, y) = x_iy_j - x_jy_i$, and the standard basis is the set of $e_i^* \wedge e_j^*$ with $i < j$. For (a)

$$\omega = e_1^* \wedge e_3^* + 2e_3^* \wedge e_1^* - 3e_2^* \wedge e_3^* = -e_1^* \wedge e_3^* - 3e_2^* \wedge e_3^*,$$

and for (b)

$$\omega = 7e_1^* \wedge e_2^* + 2e_2^* \wedge e_3^*.$$

These are linear combinations of alternating 2-tensors on $\mathbb{R}^4$, so they must also be alternating 2-tensors on $\mathbb{R}^4$, since $\mathcal{A}^2(\mathbb{R}^4)$ is a linear subspace of $\mathcal{L}^2(\mathbb{R}^4)$. $\square$

Problem 3. Consider the differential 1-forms on $\mathbb{R}^3$ given by

$$\omega = xydx + 3dy - yzdz, \quad \eta = xdx - yz^2dy + 2xdz.$$

Verify by direct computation that

$$d(d\omega) = 0, \quad d(\omega \wedge \eta) = d\omega \wedge \eta - \omega \wedge d\eta.$$

Solution. Writing $d\omega = \omega_1dx + \omega_2dy + \omega_3dz$ (so $\omega_1 = xy$, $\omega_2 = 3$, $\omega_3 = -yz$), we have

$$\begin{aligned} d\omega &= d\omega_1 \wedge dx + d\omega_2 \wedge dy + d\omega_3 \wedge dz \\ &= (ydx + xdy) \wedge dx + 0 \wedge dy + (-zdy - ydz) \wedge dz \\ &= -xdx \wedge dy - zdy \wedge dz. \end{aligned}$$

Then we can compute

$$d(d\omega) = -dx \wedge dx \wedge dy - dz \wedge dy \wedge dz = 0.$$

We also see that

$$\omega \wedge \eta = (-xy^2z^2 - 3x)dx \wedge dy + (2x^2y + xyz)dx \wedge dz + (6x - y^2z^3)dy \wedge dz,$$

so

$$d(\omega \wedge \eta) = (-2xy^2z - 2x^2 - xz + 6)dx \wedge dy \wedge dz.$$

On the other hand,

$$d\eta = 2dx \wedge dz + 2yzdy \wedge dz,$$

which gives us

$$d\omega \wedge \eta - \omega \wedge d\eta = (-2x^2 - xz)dx \wedge dy \wedge dz - (-6 + 2xy^2z)dx \wedge dy \wedge dz,$$

which is equal to $(-2xy^2z - 2x^2 - xz + 6)dx \wedge dy \wedge dz = d(\omega \wedge \eta)$ as desired. $\square$

Problem 4. Let ω be a k -form on an open set $U \subseteq \mathbb{R}^n$. We say ω **vanishes** at $x \in U$ if ω_x is the zero tensor.

(a) Show that if ω vanishes for all x in a neighborhood of x_0 , then $d\omega$ vanishes at x_0 .

(b) Give an example to show that if ω vanishes at x_0 then $d\omega$ need not vanish at x_0 .

Solution. (a) Let $\omega = \sum_I \omega_I dx_I$ where the sum is over all sequences $I = (i_1, \dots, i_k)$ of integers satisfying $1 \leq i_1 < \dots < i_k \leq n$, dx_I is shorthand for $dx_{i_1} \wedge \dots \wedge dx_{i_k}$, and the ω_I are smooth functions $U \rightarrow \mathbb{R}$. The vanishing condition means that $\omega_I(x) = 0$ for all x in a neighborhood of x_0 , so the derivative of ω_I is 0 in a neighborhood of x_0 (in particular the partials are all zero in a neighborhood of x_0). This means that, for x in this neighborhood,

$$d\omega_x = \sum_I d\omega_I(x) \wedge dx_I = \sum_I 0 \wedge dx_I = 0.$$

(b) Take $\omega: \mathbb{R} \rightarrow \mathbb{R}$ to be the 0-form given by $f(x) = x$ for all x . Then $\omega_0 = 0$, but $d\omega = dx \neq 0$ for all x . $\square$

Problem 5. We say that $\omega \in \Omega^k(U)$ is **exact** if there exists $\eta \in \Omega^{k-1}(U)$ such that $\omega = d\eta$, and we say that ω is **closed** if $d\omega = 0$. Note that $d(d\eta) = 0$ implies all exact forms are closed.

(a) Show that the 1-form

$$\omega = \frac{x}{x^2 + y^2} dx + \frac{y}{x^2 + y^2} dy$$

on $\mathbb{R}^2 \setminus \{(0, 0)\}$ is both closed and exact.

(b) Show that the 1-form

$$\omega = \frac{-y}{x^2 + y^2} dx + \frac{x}{x^2 + y^2} dy$$

on $\mathbb{R}^2 \setminus \{(0, 0)\}$ is closed.

Solution. (a) It suffices to show that ω is exact. Indeed, noting $\omega = d\eta$ where η is the differential 0-form $\ln(\sqrt{x^2 + y^2})$ proves exactness.

(b) This is just a direct derivative computation: we have

$$\begin{aligned} d\omega &= \left(\frac{-y \cdot -2y}{(x^2 + y^2)^2} - \frac{1}{x^2 + y^2} \right) dy \wedge dx + \left(\frac{x \cdot -2x}{(x^2 + y^2)^2} + \frac{1}{x^2 + y^2} \right) dx \wedge dy \\ &= \left(\frac{y^2 - x^2}{(x^2 + y^2)^2} \right) (dy \wedge dx + dx \wedge dy) \end{aligned}$$

which obviously equals zero. $\square$